On Automating Finger-Cap Array Synthesis with Optimal Parasitic Matching for Custom SAR ADC

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ABSTRACT

Due to its excellent power efficiency, the successive-approximation-register (SAR) analog-to-digital converter (ADC) is an attractive design choice for low-power ADC implements. In analog layout design, the parasitics induced by interconnecting wires and elements affect the accuracy and performance of the device. Due to the requirement of low-power and high-speed, series of very small lateral metal-metal capacitor units are usually adopted as the architecture of capacitor array. Besides power consumption and area reduction, the parasitic capacitance would significantly affect the matching properties and settling time of capacitors. This work presents a framework to synthesize good-quality binary-weighted capacitors for custom SAR ADC. Also, this work proposes a parasitic-aware ILP-based weight-dynamic network routing algorithm to generate a layout considering parasitic capacitance and capacitance ratio mismatch simultaneously. The experimental result shows that the effective number of bits (ENOB) of the layout generated by our approach is comparable to or better than that of manual design and other automated works, closing the gap between pre-sim and post-sim results.

KEYWORDS

parasitic effect, capacitance matching, placement, routing, common-centroid, analog-to-digital converter, linear programming

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1 INTRODUCTION

In recent years, the intelligence of IoT devices has brought enormous convenience into our life. Conventional AI (Artificial Intelligence) applications mostly perform data analysis and processes on the cloud. It will add a hefty burden to the servers with receiving and processing such much data. The new AIoT (AI of Things) framework holds the AI edge devices to process the data with basic neural network computing capability, so the server can only work on data computing. In order to implement such enormous neural network computing in an edge device, memory-centric designs are widely applied in AI edge devices [2]. The memory-centric designs replace the conventional computing unit with mixed memory computing unit that can achieve efficient in-memory computation (IMC) while avoiding massive amounts of data movement by deploying computation to storage. The apparent advantages and concerns of the IMC schemes [1] have been shown when applying different memory technologies, from conventional SRAM, DRAM to emerging ReRAM. However, most convenient AI edge applications such as sensors, smartphones, or even glasses, are usually designed as portable devices which are limited by their battery life. To handle such enormous data processing with a long using time is the goal for all AI edge applications. Therefore, most AI edge devices focus on lowering power consumption for long-lasting battery life.

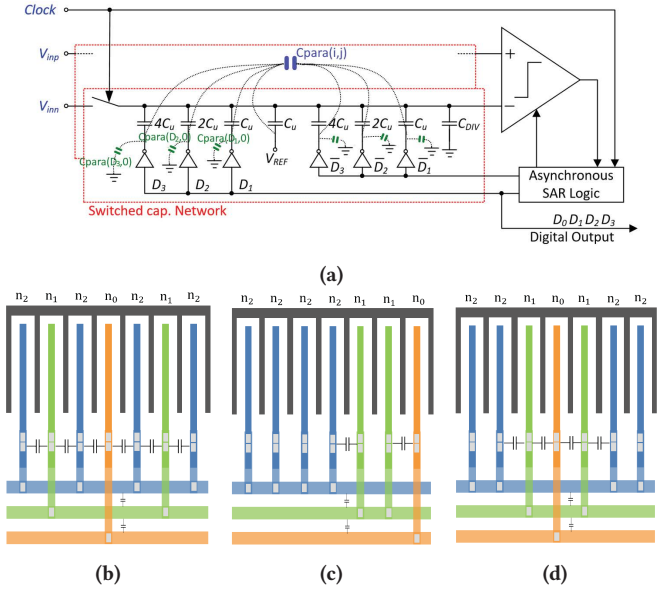


Figure 1: (a) The figure shows the SAR ADC which is used in [2] as an operation of binary search, where the blue and green capacitors are the parasitic capacitors caused by the coupling effects from interconnecting wires. Varied Finger-Cap-Array placement methods for lower power result in different parasitic capacitances, as shown in (b) fully interdigitated, (c) non-interdigitated, and (d) partially interdigitated.

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Due to its excellent power efficiency, the successive-approximation register (SAR) analog-to-digital converter (ADC) is an attractive design choice for low-power ADC implements [13, 15]. Fig. 1a is an example that performs a binary search in conjunction with SARs. It consists of two arrays of individually switched binary-weighted capacitors, a comparator, and SAR logic. Recently, quite a few works [4, 18] reported the low-power SAR ADC circuits to achieve better power efficiency. The benefit of low-power and the existence of a large number of sensor devices, especially the memory-centric AI edge applications in recent years [2], making the widespread usage of ADCs appealing in many low-power applications [14].

The binary-weighted capacitors occupy most of the power consumption in the SAR ADC because of the enormous unit capacitance and parasitic capacitance. To decrease the capacitance for reducing power consumption of the driver circuits, some works [6, 7, 10, 19] have improved the traditional metal-oxide-metal (MOM) capacitor array by adopting the structure, as Fig. 1b shows (proposed in [6]), composed of a series of lateral metal-metal capacitor units. It uses a very small capacitor to sample input voltage and compares it with the voltage generated by the charge scaling capacitor array, which is an array of individually switched binary-weighted capacitors. With these small capacitor units, the capacitor array can significantly reduce the total capacitance, layout area and charging/discharging time. This type of MOM unit capacitor array will be referred as Finger-Cap-Array structure in the rest of this work.

According to [11] and [10], the accuracy and performance of a charge-scaling DAC or a SAR ADC have a high correlation with the precision of capacitance ratio of binary-weighted capacitors, while the power consumption and settling time of a charge-scaling DAC or a SAR ADC are based on the actual capacitance values. However, the enormous parasitic capacitors induced by interconnecting wires significantly affect the matching properties among the binary-weighted capacitors when designing the capacitor layout of a charge-scaling DAC or a SAR ADC. As the blue and green capacitors shown in Fig. 1a, there exist many parasitic capacitors that are generated by the coupling effects from interconnecting wires. Moreover, [19] and [10] mentioned that the accuracy of the capacitance ratios would be further significantly affected by the parasitic capacitance from interconnecting wires and placement style unless we reduce the capacitance of the unit capacitor. The parasitic capacitance results in a longer settling time, mismatch of capacitance ratio, and higher switching power. Therefore, a simple and effective approach is necessary to minimize the impact on circuit accuracy due to the coupling effect.

1.1 Previous Works on Binary-Weighted Capacitor Layout Synthesis

Although the techniques of ratioed capacitors layout synthesizing had been extensively studied, most of the previous works focus on the placement and/or routing in the traditional MOM-CAP structure. [5, 16] emphasized the matching properties of common-centroid placement or even considering the influence from the gradient mismatch. However, they ignored the parasitic capacitance caused by the routing which may destroy the resulting matching properties of the ratioed capacitors. [9, 11] proposed some dispersive and common-centroid routing methods and guidelines for unit capacitor array to minimize the parasitic capacitance and/or the mismatch of the capacitance ratio. Nonetheless, the impact from routing parasitics becomes enormous with the tiny unit capacitor in Finger-Cap-Array. None of them can treat the matching properties in this structure.

[6] first proposed the structure of Finger-Cap-Array, which is also used in our SAR ADC for further lower power consumption. A few recent works discussed some guidelines or methods for designing this kind of binary-weighted capacitors structure. [12] proposed a layout method that used the structure of Finger-Cap-Array with an equal pitch to avoid routing complexity and keep the regularity. [10] proposed a placement style of partially interdigitated common-centroid SAR ADC layout that strikes a balance between the inter-bit parasitic capacitance and systematic mismatch.

With the necessary of power efficiency, switched binary-weighted capacitors with the structure of Finger-Cap-Array has been widely used in many low-power SAR ADC circuits. However, none of the previous works optimizes the matching properties in the structure of Finger-Cap-Array. Although Lin et al. [11] proposed an optimizing algorithm that considers the total parasitic and matching properties of the ratioed capacitors with simultaneously routing and placement, the routing area is too scant to optimize the capacitance ratio matching in the structure of the Finger-Cap-Array.

1.2 Our Contribution

This work presents a new framework to synthesize a Finger-Cap-Array layout for low energy asynchronous SAR ADC. Compared with the previous works in Section 1.1, we list the features of our work as follows.

(1) We first adopt the common-centroid placement style, "partially interdigitated" (PI) in [10], to automatically generate an optimal binary-weighted capacitors placement result in SAR ADC with balanced parasitic capacitance and capacitance ratio mismatch. Compared with the previous works applying the traditional MOM-CAP structure, the complexity is greatly decreased with the PI placement in the Finger-Cap-Array structure.
(2) In the routing stage, we present the parasitic-aware ILP-based weight-dynamic network routing algorithm to generate the optimal layout. This is the first work that implements the routing problem considering parasitic capacitance and matching properties of the ratioed capacitors in the structure of Finger-Cap-Array. This routing approach can concurrently route a set of interfering nets and obtain the optimal routing solutions.
(3) Our approach helps designers save much time when designing the switched binary-weighted capacitors. The impact of capacitance-ratio mismatch was identified after post-layout simulation, leading to expensive re-design iterations by inspecting the physical coupling of interconnection concerning one another. Based on the ILP-based algorithm, we can obtain a routing result with minimized device mismatch.

The remainder of this work is organized as follows. Section 2 presents the problem formulation of automating layout generation for common-centroid binary-weighted capacitors. Sections 3 and 4 introduce our two-stage method, including parasitic-aware common-centroid capacitor placement and ILP-based weight-dynamic network routing algorithms. The experimental results are shown in Section 5. Finally, Section 6 summarizes our work.

2 PROBLEM FORMULATION

The primary purpose of our work is to minimize the total parasitic capacitance (C) and capacitance ratio mismatch (M) of each net simultaneously. Thus, the placement and routing problem in our work is described as follows.

We are given a unit finger-capacitor and a set of nets with corresponding capacitor ratios. First, we should find a common-centroid placement result considering parasitic capacitance for the following routing step. With the placement result, the target is to find nonoverlap routing paths that connect the IO-pin and all unit capacitors that belong to the same net considering the total parasitic capacitance and capacitance ratio mismatch caused by the paths. It can be transferred to minimizing the objective function:

$$\varphi = \sum_{i}^{N} \alpha * \widetilde{C}_i + \beta * \left| \widehat{C}_i - \widetilde{C}_i \right| \quad (1)$$

where α and β are user-defined variables and N is the set of nets. $\widetilde{C}_i$ and $\widehat{C}_i$ represent the real and targeted capacitance of net i in the binary-weighted capacitors, respectively. The capacitance values induce the parasitic capacitance estimated by our parasitic model from interconnection, as seen in Fig. 1a.

With the minimized φ, we can get the optimized binary-weighted capacitor layout with a good balance between parasitic capacitance and matching.

3 COMMON-CENTROID PLACEMENT OF FINGER-CAP-ARRAY

In this section, we introduce the placement method of our work. The objective of this stage is to generate the optimal placement result for the follow-up routing stage. In the layout of a Finger-Cap-Array, any tiny change of the location will affect the mismatch of the capacitance ratio. Thus, the placement problem includes two parts: the one-dimension finger unit capacitor placement and I/O pins placement.

3.1 One-Dimension Unit Finger Capacitors Placement

The Finger-Cap-Array layout employing the comb structure [7] places the unit finger capacitors in a one-dimension array, as illustrated by an example of 4-bit ADC in Fig. 1b where the capacitance ratio of n_0, n_1, and n_2 is $1:2:4$. The horizontal wires are the bit lines that connect with the I/O pins, and the vertical wires connect the horizontal bit lines with their unit capacitors. In order to avoid systematic mismatch due to the unexpected parasitic capacitance, the horizontal bit lines usually are placed far away from the unit capacitors. Moreover, the conventional layout distributes the binary-weighted capacitors with common-centroid form, as indicated by the naming n_i in Fig. 1b, to avoid the process gradient. However, as mentioned in [10], this "fully interdigitated" (FI) layout results in significant parasitic capacitance between the vertical bit lines. We can only reduce the parasitic capacitance between vertical lines by changing the placement order of unit capacitors because the vertical line spacing is set by the unit capacitor pitch which is very small. For instance, n_2 sees six parasitic capacitors (C_{para}) to the vertical lines of n_1 and n_0. Since n_2 remains at a DC voltage when n_3 switches, the total parasitic capacitance to the ground will be increased by these parasitic capacitors. It results in a longer settling time and higher switching power.

A solution is to reduce the capacitors between vertical lines: to group the unit capacitors belonging to the same bit/net together. As the example shown in Fig. 1c, n_3 only sees one C_{para} to the other vertical bit lines. We can significantly reduce the inter-bit parasitic capacitance with this "non-interdigitated" (NI) layout. However, NI layout costs a higher mismatch from process gradient since it lost the structure of common-centroid.

In this work, we adopt the "partially interdigitated" (PI) [10] as shown in Fig. 1d that strikes a balance between the inter-bit parasitic capacitance and common-centroid matching. The PI layout places the groups of the unit capacitors belonging to the same bit symmetrically to the center of the array. This technique preserves the common-centroid structure as well as the total parasitic capacitance. In addition, [3] mentioned that the random mismatch from the process variation decreases as the unit capacitors of each group is placed dispersed. For high-bit SAR ADC, although there are many other feasible ways to place the one-dimensional unit finger capacitors, designers often discourage the use of high-bit SAR ADC in IMC devices because of its higher mismatch. And for low-bit SAR ADC, the PI placement style shows the best result compared with other placement styles.

To generate the optimal placement result of the unit capacitor array automatically, we first sort the nets as n_0,...,n_{n-1} and assign the unit capacitors with the capacitance ratio r_i, where r_i is the capacitance ratio of n_i and $r_0 < r_1 < ... < r_{n-1}$. The net with capacitor ratio r_i will be assigned r_i unit capacitors. Then we place the unit capacitors symmetrically to the center of the array as the PI structure. To minimize the random mismatch from the process variation, the unit capacitors of the net with the smallest (biggest) capacitance ratio will be placed at the center (outermost) of the array. With the result of this step, we can get the unit capacitor array placement with a good balance between the inter-bit parasitic capacitance and common-centroid matching.

3.2 I/O-pin Placement

After finishing the unit capacitor array placement, we should place the I/O pin for the follow-up routing stage. This stage is to determine the order of the horizontal bit lines. Since the parasitic capacitance of a net should also match its capacitance ratio, the net with a higher capacitance ratio needs more parasitic capacitance. As the result, the order of the horizontal bit lines will be the same as the order of capacitor ratio. The horizontal bit line of the net with the highest ratio will be placed at the bottom, which is the farthest horizontal wire from the unit capacitor array. With this strategy, the net will get more parasitic capacitance from the longer vertical wires. Fig. 2 shows a result of an example with Net n_0-n_2 with capacitance ratio 1:2:4. The unit capacitors are placed in PI common-centroid form to balance the inter-bit parasitic capacitance and common-centroid matching. And with the placement of the IO pins illustrated as black rectangles, the horizontal wires are placed in the order of n_0, n_1, n_2 from top to bottom.

4 ILP-BASED ROUTING CONSIDERING PARASITIC MATCHING

This section describes the detail of our routing flow to complete the interconnect of the unit capacitors and I/O pins in the Finger-Cap-Array structure. The routing stage should consider the parasitic capacitance caused by the coupling effect from interconnection.

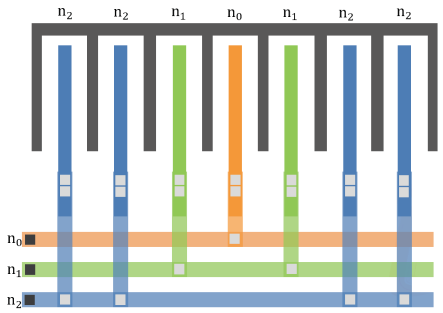


Figure 2: An example of a 4-bit ADC Finger-Cap-Array layout result in PI form with the placement strategies. The strategies will induce the routing parasitics ratio matching the capacitance ratio.

Each choice of paths/wires will result in different parasitic capacitance, which significantly affects the accuracy and performance of ADC/DAC. Thus, we apply the parasitic-aware ILP-based network routing algorithm to complete the connection of all nets simultaneously. By transferring the layout into a weight-dynamic network, we can obtain an optimal routing result by implementing the cost minimization problem with an ILP-based algorithm.

In the routing stage, we first transfer all routable paths into a network graph. Then, the proposed ILP-based routing is implemented to select a set of paths for completing the connection with consideration of parasitic capacitance and capacitance ratio mismatch.

4.1 Network Construction

In the first step of the routing stage, we build a network graph $G(V, E)$, where V is the set of vertices consist of all I/O pins, unit capacitor pins and virtual points for connecting horizontal wires with all vertical wires in the same net. And E is the set of edges consist of all vertical and horizontal wires in all routable layers. Fig. 3 shows an example of our network formulation. Fig. 3a shows the 2D connection of the placed result. The connection shows all routable paths we determined, including the following three types:

- A horizontal path extended from the I/O pin for each net.
- A vertical path for each unit capacitor to connecting the unit capacitor pin with the horizontal path.
- A path for each adjacent unit capacitor pins belonging to the same net.

Fig. 3b shows the corresponding network graph of the placed layout. There are three routable layers ($M1$, $M3$, $M5$) and ($M2$, $M4$, $M6$) for the horizontal and vertical paths. For simplicity, we only show nets n_0 and n_1. Note that since all pins of the unit capacitor in the same net should be connected to the same horizontal wire, the virtual vertex, $V_H^{n_1}$, is created to help n_1 connect its all vertical wires with the horizontal wires.

4.2 Parasitic-aware ILP-based Weight-Dynamic Network Routing

Under our network construction, this step is going to find a set of edges that finishes the interconnection of the nets. At the same time, the result should present the minimum cost of the objective function (1) for obtaining the optimal layout with minimized parasitic capacitance and capacitance ratio mismatch in equation (1).

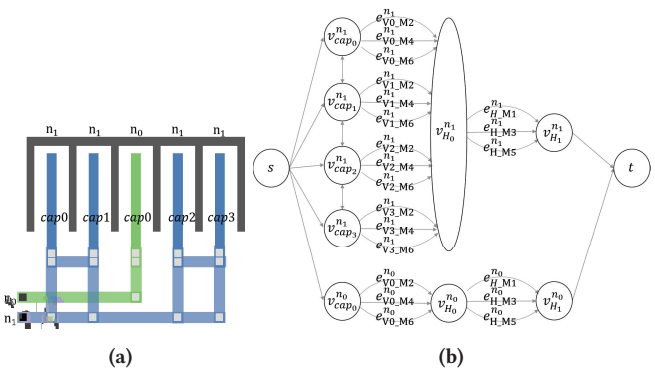


Figure 3: An example of (a) the 2D connection of placed result and (b) the corresponding network graph with two nets, n_0 and n_1, and a capacitance ratio of $1:4$.

Table 1: Notation description for ILP-based routing.

Notation	Meaning
N	set of nets
n_i	i^{th} net of N
r_i	capacitance ratio of n_i
$\widehat{C}_i$	target capacitance of n_i
$\widetilde{C}_i$	real capacitance of n_i
C_u	capacitance of a unit capacitor
$Cpara_i$	total parasitic capacitance from interconnection of n_i
e_k^i	k^{th} edge of n_i
GP_E	set of edge groups for guaranteeing connectivity

In the routing problem of Finger-Cap-Array layout, each choice of edges will affect the cost of other nets because of the coupling effects. In other words, the weight of each edge will change with different results of edges choosing. [8] named this network as weight-dynamic network which is a dynamic graph with only edge-weight changes. To get an optimal routing result, we implement the cost minimization problem with ILP to select a set of edges with minimum cost from the dynamic-weighted edges for simultaneously minimizing parasitic capacitance and capacitance ratio mismatch.

In objective function (1), $\widetilde{C}_i$ and $\widehat{C}_i$ are calculated by the following formulations. As shown in equation (2), $\widehat{C}_i$ is determined as the r_i times base capacitance. The base capacitance depends on the total capacitance of n_0, which has the lowest capacitance ratio in all nets sorted in Section 3. As well as $\widetilde{C}_i$ is calculated by summing of the unit capacitances (C_u) and the total interconnect parasitic capacitance of n_i ($Cpara_i$). Note that the $Cpara_i$ are calculated with a prepared look-up table. Different routing layers derive different capacitance values. Also, there are different coupling capacitance values between different pairs of interconnecting wires. The various capacitance values were pre-stated in the look-up table for the ILP model.

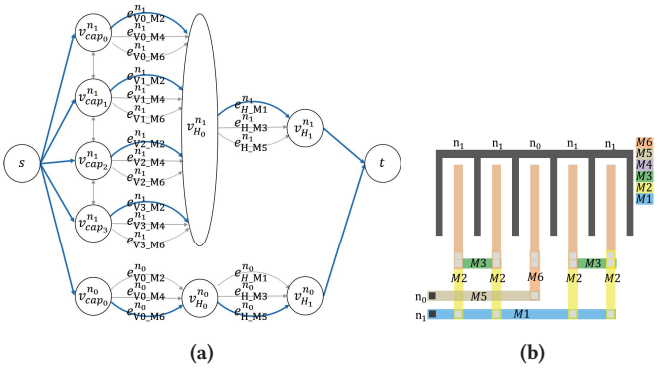
$$\widehat{C}_i = \widehat{C_0} * r_i / r_0, \text{ and } \widetilde{C}_i = 2^{r_i - 1} * C_u + Cpara_i, \quad (2)$$

Finally, in order to guarantee the connectivity of all nets, we propose the following constraint functions for the ILP model. We first group the edges that start from and end at the same vertices as a gp_E in the group set GP_E. Also, we group the vertical edges that connect

to the adjacent unit capacitors belonging to the same net as a node to represent the connectivity. The connectivity can be guaranteed by choosing at least one edge for each gp_E as shown in the constraint function (3):

$$\sum_{e}^{gp_E} |e| \geq 1,\ gp_E \in GP_E \tag{3}$$

After modeling the objective function and the constraints in this section, we can obtain a set of edges that presents the optimal layout with the ILP-solver. Fig. 4 shows the result simultaneously considering parasitic capacitance and capacitance ratio mismatch with the example shown in Fig. 3a. The ILP-based routing algorithm minimizes the mismatch of capacitance ratio by choosing the vertical wires in the M2 layer and picking more edges to increase the parasitic capacitance of n_1 for matching the ratio of the total capacitance. With the tradeoff, we can obtain an optimal layout with a better simulation result. Finally, we transform the synthesized layout into the Cadence skill language for further verification and post-simulation.



(a) (b)

Figure 4: (a) The edge-selection result and (b) the corresponding layout result of the example shown in Fig. 3a. The bolded edges are the flow edges selected by the ILP-solver with minimized cost.

5 EXPERIMENTAL RESULT

The proposed flow and methods for Finger-Cap-Array layout synthesis are implemented in C++ programming language on Intel Exon E5-2630 machine with a 2.2 GHz 64-bit CPU. The experimental results show the advantage of our approach compared with the manual layout and the recent works [11] [9] which are implemented by ourselves. Note that [11] and [9] only consider the binary-weighted capacitor layout optimization in the traditional MOM capacitor array structure. We use two SAR ADCs, including 4-bit [2] and 8-bit [17], as our benchmark circuits. We generated the layouts of binary-weighted capacitors based on different approaches with leading-edge foundry 28nm technology. Then we performed post-layout parasitic extraction and simulation to evaluate the ENOB for the two SAR ADCs. Note that we can get the optimized layout from our approach in the runtime of 0.01 seconds since the runtime of the ILP algorithm depends on the number of unit capacitors and nets. Because of the small bits of these benchmarks, all of the automated generation techniques can generate the result within a very short runtime. So it is not necessary to compare the runtime for each automated generation technique.

First, we compared our approach with the manual layout and the recent approaches [11] [9]. Since [11] and [9] did not consider the structure of Finger-Cap-Array, we applied the same placement resulting from our approach to their approaches. Table 2 shows the result of our experiments, which compares the total capacitance ("Total cap."), capacitance ratio mismatch ("cap-ratio Mismatch"), and the ENOB of the SAR ADCs after performing post-layout parasitic extraction and simulation. The column "Comparison" is the improving ratio of ENOB calculated as $(A - B)/(Ideal - B)$, where A and B are the ENOB of the approach and the manual layout, and Ideal ENOBs for the 4-bit and 8-bit SAR ADC are 4 and 8 bits.

According to Table 2, each of [11], [9] and our approach reduces the total parasitic capacitance and capacitance ratio mismatch comparing with the manual layout. Each approach optimized the ENOB by reducing the total parasitic capacitance and capacitance ratio mismatch. However, [11] and [9] could not make good use of their advantage in the structure of the Finger-Cap-Array due to the very small routing space. Because the vertical and horizontal wires are the only space we can route in this structure, our approach can achieve a better ENOB closer to the post-simulation with much less total parasitic capacitance and capacitance ratio mismatch compared with [11], [9]. Our approach can result in less total parasitic capacitance and capacitance ratio mismatch comparing because we further consider the parasitic capacitance caused by the horizontal and vertical wires while [11] and [9] did not. Note that the total capacitance of our result in 8-bit SAR ADC is larger than [11] and [9]. It shows the effectiveness of our approach to obtain the layout with a smaller parasitic ratio mismatch and better ENOB by appropriately utilizing the interconnect parasitic capacitance. In addition, the utilized parasitic capacitance does not affect the power performance too much because the adoption of very small unit capacitors has significantly reduced the total capacitance. Moreover, the mismatch of a SAR ADC ("Cap-Ratio Mismatch") represents the maximum net capacitance ratio mismatch in all nets, while the average of it means the average capacitance ratio mismatch of all nets.

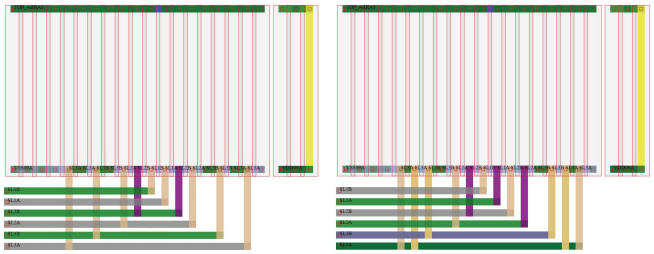
Finally, we compared the proposed approaches with and without considering capacitance ratio mismatch for both the 4-bit and 8-bit SAR ADCs. According to the results shown in the last two rows in Table 2 and the layout shown in Fig. 5, although our approach only considering parasitic capacitance can further reduce the total capacitance of the layout, the capacitance ratio mismatch is very large. We can obtain a layout with a better post-simulation result if we can find a good balance between parasitic capacitance and capacitance ratio mismatch.

6 CONCLUSIONS

Due to the increasingly advanced manufacturing process technology, the integrated circuit density increases, increasing the influence of coupling effects on the circuit performance. The importance of automatic placement and routing tools for analog layout is thus out of the question. In this work, we present a framework to synthesize binary-weighted capacitors in the structure of Finger-Cap-Array. The partially interdigitated common-centroid placement is presented to generate a placement result considering the parasitic effect and the gradient mismatch. Next, the parasitic-aware ILP-based network routing algorithm is performed to guarantee the connectivity for all nets simultaneously as well as minimize the parasitic capacitance and capacitance ratio mismatch with the proposed objective function and constraints. The experimental result shows that

Table 2: Comparison of total capacitance, interconnect parasitic capacitance, capacitance ratio mismatch, and ENOB of 4-bit and 8-bit SAR ADC for manual layout, [11], [9], and our approach

4-bit SAR ADC					
Approach	Total cap.	Interconnect C_{para}	Cap-Ratio Mismatch	ENOB (preSim: 3.4293)	Comparison
Manual	2.06e-14 fF	9.37e-15 (45.5%)	32.9% (avg=12%)	3.263 bits	–
Lin et al.'s approach [11]	1.83e-14 fF	7.3e-15 (39.4%)	30.2% (avg=9.63%)	3.374 bits	15.1%
Ho et al.'s approach [9]	1.97e-14 fF	8.46e-15 (43%)	24.2% (avg=8.5%)	3.342 bits	10.8%
Our approach	1.79e-14 fF	6.7e-15 (38.4%)	9.96% (avg=3.46%)	3.429 bits	22.5%
Our approach only considering total cap.	1.67e-14 fF	5.51e-15 (33%)	34.2% (avg=13.3%)	3.363 bits	13.6%
8-bit SAR ADC					
Approach	Total cap.	Interconnect C_{para}	Cap-Ratio Mismatch	ENOB (preSim: 7.3132)	Comparison
Manual	1.26e-13 fF	1.172e-14 (9.3%)	46% (avg=25.3%)	5.882 bits	–
Lin et al.'s approach [11]	1.33e-13 fF	4.674e-14 (35.14%)	31.2% (avg=17.6%)	6.156 bits	12.9%
Ho et al.'s approach [9]	1.48e-13 fF	3.372e-14 (22.7%)	39.1% (avg=19.2%)	5.978 bits	4.5%
Our approach	2.31e-13 fF	1.167e-13 (50.5%)	21.9% (avg=11.2%)	6.724 bits	39.8%
Our approach only considering total cap.	1.25e-13 fF	1.046e-14 (8.39%)	56% (avg=27.6%)	5.464 bits	-19.7%



(a) (b)

Figure 5: A 4-bit SAR ADC binary-weighted capacitors layout (a) without and (b) with considering capacitance ratio mismatch. The parallel wires in (a) try to stay away from each other with staggered layer-assignment and adjacent avoiding. And some parallel wires in (b) try to get closer for more parasitics to match the capacitance ratio.

our approach's ENOB of the layout is comparable with manual design and other automated research.

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